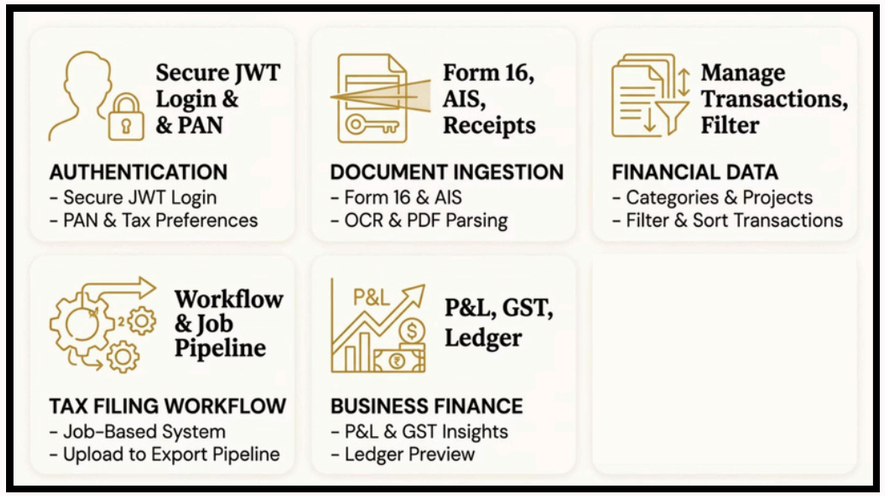
TaxHacker

YourAI-Powered FinancialOperating System

Team Name : ByteCoders



The Problem We Solve

Today, individualsandbusinessesstrugglewithafragmented,time-consuming financial workflow. TaxHacker eliminates every pain point inone unifiedplatform.

ManualDataEntry

Tracking expenses across apps, receipts, and spreadsheets is tedious and error-prone.

ComplexTaxRules

Understanding deductions, regimes, and ITR forms requires expert knowledge most users lack.

Missed Savings

Deductions and tax-saving opportunities go unnoticed, costing users money every year.

HoursWasted

Preparing tax filings manually consumes hours that could be spent on productive work.

What Makes TaxHacker Stand Out

"TaxHacker is an AI-powered financial operating system that automates tax filing, tracks expenses, and provides personalised financial intelligence."

End-to-End Automation

UploadtoITR—zeromanualintervention required at any stage.

Custom AI Fields

User-definedextractionrules — a rare and powerful capability in the market.

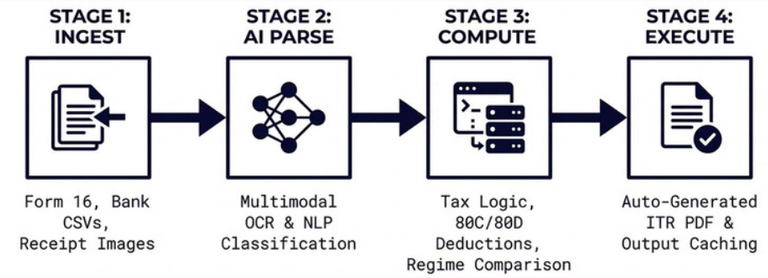
Multimodal AI

Processesimages,CSVs — handles any document format seamlessly.

Tax Ecosystem

Notjustataxtool—Taxintelligence and advisory support built in.

- **Income Detection**
  Identifiessalary, interest,freelance, and other income sources from uploaded documents.
- **Deduction Parsing**
  Automaticallydetects80C,80D, HRA, and other eligible deductions from financial data.
- **Tax Computation**
  Calculates gross income, taxable income, tax liability, TDS credits, and refund/payable amount.
- **Regime Comparison**
  Compares Old vs New tax regime and recommends the optimal choice — e.g., "Save ₹X using new regime."



Tech Stack & Architecture

Frontend & API

- **React.js** — Responsive frontend
- **FastAPI** — High-performance API layer
- **JWT** — Secure authentication

Data Layer

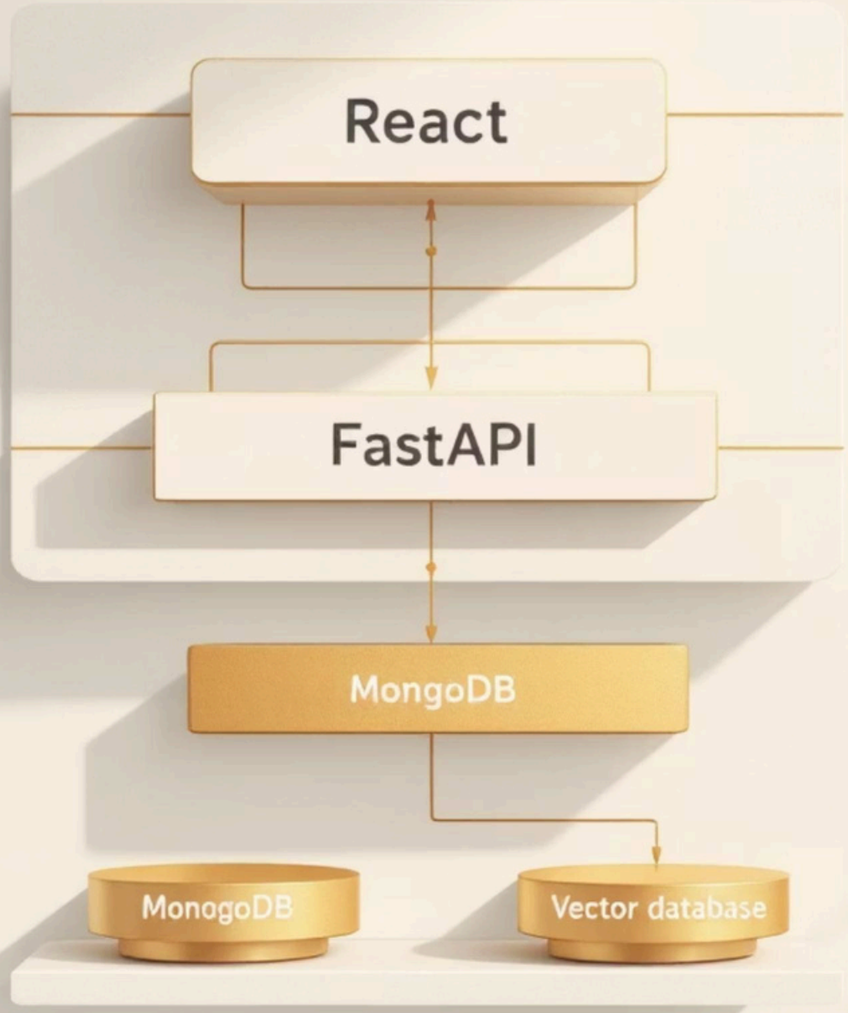
- **MongoDB** — Primary data storage
- **Vector DB** — RAG-powered knowledge retrieval
- **Pageless Index** — Efficient document indexing

AI Layer

- **GroqLLM** — Fast language model inference
- **LangChain** — AI orchestration & RAG pipeline
- **OCR Engine** — Image-to-text recognition

Flow

React.js → FastAPI → MongoDB + Vector DB → AI Layer (OCR + LLM + LangChain)



Challenges Faced

1 Database Selection Issue

Initially used SQLite, but realized it is not suitable for production deployment on platforms like Render due to data persistence issues. Migrated to PostgreSQL for reliability and scalability.

2 Backend Framework Limitation

Started with Flask, but faced limitations due to lack of native async support. Switched to FastAPI for better performance and asynchronous handling.

3 Data Privacy & Security

Ensured secure handling of sensitive financial data by sending only transaction rows to the LLM while excluding personal information (Name, Account Number, PAN, etc.), reducing risk of data exposure.

4 Unstructured Data Handling

Bank statements had inconsistent formats. Used LLM to convert transactions into a structured, uniform table format for consistent processing and analysis.

Future Scope & Scalability

1

Direct Government Portal Integration

- Automated filing with Income Tax Department APIs
- One-click submission of ITR
- End-to-end compliance automation

2

Multi-Country Tax Support

- Extend system beyond India
- Modular tax engine for different countries
- Global user base expansion

3

Advanced Financial Intelligence

- Real-time expense tracking via bank integrations
- Predictive analytics:
- future tax liability

4

Smarter AI Personal Advisor

- Deep personalization using user behavior
- Investment suggestions (ELSS, FD, mutual funds)
- Long-term financial planning